

## **22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA**

### **S2 B.Tech CSE-AI**

**Department of Computer Science and Engineering**

**Amrita School of Computing, Amritapuri Campus**

### **CASE STUDY**

#### **PHASE – I (Analysis)**

**Problem Identification & Class Mapping (CO1)**

**Group No : 6**

**Team Members :**

| Roll No          | Student Name |
|------------------|--------------|
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- 1. Problem Title:** Banking Account Management System
- 2. Problem Description:** Manual banking operations lead to record inconsistencies, delayed transactions, and reduced efficiency.
- 3. Identified Entities (Classes)**

| Class Name  | Description                                                                   |
|-------------|-------------------------------------------------------------------------------|
| Customer    | A bank user who performs banking operations.                                  |
| Account     | Customer's bank account and store all account-related details                 |
| Transaction | Represents financial activity                                                 |
| BankStaff   | Bank employees who manage customr accounts and verify or process transactions |

#### 4. Class Attributes and Behaviour

| Class Name  | Attributes                                                                               | Methods(behaviours)                                                                                          |
|-------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Customer    | - customerId<br>- password<br>- name<br>- address<br>- phoneNumber<br>- emailId<br>- dob | + login()<br>+ accountDetails()<br>+ deposit()<br>+ withdraw()<br>+ checkBalance()<br>+ transactionHistory() |
| Account     | - accountNumber<br>+ AccountType<br>- balance                                            | + deposit(amount)<br>+ withdraw(amount)<br>+ getBalance()<br>+ updateBalance()                               |
| BankStaff   | - staffId<br>- bname<br>+ designation<br>+ department<br>- contactNumber                 | + verifyCustomer()<br>+ approveTransaction()<br>+ customerRecords()<br>+ updateCustomerDetails()             |
| Transaction | - transactionId<br>+ amount<br>+ date<br>+ time                                          | + processTransaction()<br>+ generateReceipt()<br>+ getTransactiondetails()                                   |